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#include <stdio.h>
#include <ctype.h>
#define MAX 100

char stack[MAX];
int top = -1;

// Push function
void push(char x) {
    top = top + 1;    // Move top one step up
    stack[top] = x;    // Store value at top position
}

// Pop function
char pop() {
    char value;
    value = stack[top]; // Get top element
    top = top - 1;      // Move top one step down
    return value;       // Return removed value
}

// Function to check precedence
int precedence(char x) {
    if (x == '+' || x == '-')
        return 1;
    if (x == '*' || x == '/')
        return 2;
    if (x == '^')
        return 3;
    return 0;
}

int main() {
    char infix[MAX], postfix[MAX];
    int i = 0, j = 0;
    char ch;

    printf("Enter Infix Expression: ");
    scanf("%s", infix);

    while (infix[i] != '\0') {
        ch = infix[i];

        // If operand, add to postfix
        if (isalnum(ch)) {
            postfix[j++] = ch;
        }
        // If '(', push to stack
        else if (ch == '(') {

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        push(ch);
    }
    // If ')', pop until '('
    else if (ch == ')') {
        while (stack[top] != '(')
            postfix[j++] = pop();
        pop(); // remove '('
    }
    // If operator
    else {
        while (top != -1 && precedence(stack[top]) >= precedence(ch))
            postfix[j++] = pop();
        push(ch);
    }
    i++;
}

// Pop remaining operators
while (top != -1)
    postfix[j++] = pop();

postfix[j] = '\0';

printf("Postfix Expression: %s\n", postfix);

return 0;
}

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